

 UNIVERSITY OF TECHNOLOGY - VNUHCM FACULTY OF COMPUTER SCIENCE AND ENGINEERING	HOMEWORK 4		Semester/Academic year		1	2026-2027
			Due		14/10/2026	
	Course title		Machine Learning			
	Course ID		CO3117			
	Duration		Take-home	Homework set		HW4
Notes: Handwritten homework, submitted as homework/hw4-submission.pdf with tag hw4.						

Instructions

- **Bayesian learning and Bayesian networks** (Chapters 4 and 6 (Part I)). Five problems, each worth 2 points (0.5 per part); total 10 points.
- **Write by hand**, on paper or with a stylus. Show your working: a bare number earns no credit. Put your full name, student ID and the set number at the top of the first page.
- **Work on your own.** Do not use AI tools, solution manuals or classmates' work before your submission is tagged. You may use your lecture notes and the course slides. Any later AI use goes in AI_USE.md, as for the weekly drills.
- **Submit in your course repository.** Scan all pages, in order, into one PDF of at most 5 MB, named homework/hw4-submission.pdf. Commit it, run python tools/check.py --checkpoint hw4, then git tag hw4 and git push origin main hw4.
- **Deadline: 14/10/2026 23:59** (the time GitHub receives the tag hw4). Solutions are released on 15/10/2026. Afterwards you may add homework/hw4-corrections.md; never replace or re-tag the submission.

Problem 1 (L.O.2) (2 pts): Answer the following questions about estimating the probability θ that a coin lands heads. In 10 tosses the coin shows 7 heads and 3 tails.

- Write the likelihood $L(\theta)$ and derive the maximum likelihood estimate $\hat{\theta}_{ML}$.
- Using $\hat{\theta}_{ML}$, what is the probability that the next two tosses are both heads?
- Suppose only three values are possible, $\theta \in \{0.3, 0.5, 0.7\}$, with prior probabilities 0.2, 0.6 and 0.2. The likelihoods are $L(0.3) \approx 7.50 \times 10^{-5}$, $L(0.5) \approx 9.77 \times 10^{-4}$ and $L(0.7) \approx 2.22 \times 10^{-3}$. Compute the posterior of each value and find the MAP value.
- Using the posteriors of part C, compute the probability that the next toss is heads by averaging over the three values.

Problem 2 (L.O.2) (2 pts): Answer the following questions about the Bayes optimal classifier. After training, three hypotheses have posterior probabilities $P(h_1|D)=0.45$, $P(h_2|D)=0.30$ and $P(h_3|D)=0.25$. On a new instance x , h_1 predicts + while h_2 and h_3 predict −.

- Which hypothesis is the MAP hypothesis, and what does it predict for x ?
- Write the Bayes optimal classification rule and apply it to x .
- In what sense is the Bayes optimal classifier optimal? Can the labelling it assigns to all instances differ from the labelling given by every single hypothesis in H ?
- Why is the Bayes optimal classifier rarely used directly in practice?

Problem 3 (L.O.2) (2 pts): Answer the following questions using the naive Bayes classifier on the training data below (attribute values: Outlook $\in \{\text{Sunny, Rainy}\}$, Wind $\in \{\text{Weak, Strong}\}$; class Play $\in \{\text{Yes, No}\}$).

#	Outlook	Wind	Play
1	Sunny	Weak	Yes
2	Rainy	Strong	No
3	Rainy	Weak	Yes
4	Rainy	Strong	No
5	Sunny	Weak	Yes
6	Rainy	Weak	Yes
7	Sunny	Strong	Yes

8	Rainy	Strong	No
9	Rainy	Weak	No
10	Sunny	Weak	Yes

- State the conditional independence assumption of naive Bayes and write its classification rule v_{NB} .
- Classify the new day (Outlook = Rainy, Wind = Strong). Show every probability you use.
- Classify the new day (Outlook = Sunny, Wind = Strong) without smoothing. What problem do you observe?
- Repeat part C using Laplace (add-one) smoothing for the conditional probabilities. Does the decision change?

Problem 4 (L.O.2) (2 pts): Answer the following questions about the Bayesian network with edges $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$ and $C \rightarrow E$.

- List the parents and the non-descendants (other than the parents) of each node.
- Write the local conditional independence statements that the network asserts.
- Classify each query as causal (predictive), diagnostic or intercausal reasoning: (i) $P(D | A=1)$; (ii) $P(A | D=1)$; (iii) $P(A | C=1, B=1)$.
- Factors $\phi_1(A, C)$ and $\phi_2(C, D)$ over binary variables have $\phi_1(A=1, C=1)=0.3$ and $\phi_2(C=1, D=0)=0.1$. What is the scope of the product $\psi = \phi_1 \cdot \phi_2$, how many entries does it have, and what is $\psi(A=1, C=1, D=0)$?

Problem 5 (L.O.2) (2 pts): Answer the following questions about inference in the Bayesian network $Rain \rightarrow WetGrass \leftarrow Sprinkler$, with $P(R=1)=0.2$, $P(S=1)=0.4$, and $P(W=1 | R, S)$ equal to 0.99 for $(R, S)=(1, 1)$, 0.90 for $(1, 0)$, 0.80 for $(0, 1)$ and 0.05 for $(0, 0)$.

- Write $P(R, S, W)$ as the product of the local distributions of this network, then compute $P(R=1, S=0, W=1)$.
- Compute $P(W=1)$.
- Compute $P(R=1 | W=1)$. What type of reasoning is this?
- Compute $P(R=1 | W=1, S=1)$ and compare it with part C. Name and explain the effect you observe.

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